



Unit 4 · Outcome 1 · Climate Change

The natural greenhouse effect keeps us alive. The enhanced one is the problem.

KK04

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



Start here: Earth's energy balance

Earth runs on a budget. Energy arrives from the Sun and energy leaves to space, and when the books balance, the planet's temperature holds steady. Here is the three-step story that every greenhouse answer is built on:

The single most important idea: the atmosphere is transparent to incoming shortwave but not to outgoing longwave. Sunlight sails in; the heat the surface radiates back out gets partly intercepted. That asymmetry is the entire mechanism.

An atmospheric gas that absorbs and re-emits longwave (infrared) radiation, contributing to the warming of Earth's surface. The main ones are water vapour (H_2O), carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O) and ozone (O_3).

The natural greenhouse effect

The natural greenhouse effect is the warming produced by naturally occurring greenhouse gases as they absorb outgoing longwave radiation and re-emit it in all directions. Some of that re-emitted energy heads back toward the surface, so the surface and lower atmosphere stay warmer than they otherwise would.

How much warmer? This is the figure examiners love:

That +33 °C is not a problem to be fixed. It is the reason liquid water, agriculture, and you exist. Critically, the natural effect sits in equilibrium: over a year, energy in $\approx$ energy out, so the global average temperature is roughly stable. The greenhouse gases involved cycle naturally — CO_2 from respiration, volcanoes and decay; methane from wetlands; water vapour from evaporation.

The warming of Earth's surface caused by naturally occurring greenhouse gases absorbing and re-emitting longwave radiation, raising the global average temperature from about -18 °C to about +15 °C. It is essential for life and operates at a stable equilibrium.

The enhanced greenhouse effect

The enhanced greenhouse effect is the additional warming that results when human activities raise greenhouse-gas concentrations above natural levels. More greenhouse gas molecules means more outgoing longwave radiation is absorbed and re-emitted back toward the surface, so less heat escapes to space. Energy in now exceeds energy out — the budget no longer balances — and the planet's average temperature rises until a new, warmer equilibrium is reached.

At Cape Wirreanda, the observatory's CO_2 record shows two things at once: a small seasonal wiggle (the natural breathing of the biosphere) riding on a relentless upward climb (the human signal). The wiggle is the natural cycle; the climb is the enhanced effect being written into the air in real time.

The additional warming of Earth's surface caused by the increased concentration of greenhouse gases from human activities (e.g. fossil-fuel combustion, agriculture, land-use change), which traps more outgoing longwave radiation and produces a net energy imbalance and rising global temperatures.

The differences — the exam table

The enhanced effect is not a separate or new phenomenon. It is the intensification of the natural effect by extra human-added gases. An answer that treats them as two different mechanisms has missed the point of the dot point.

Say it like this: "The enhanced greenhouse effect uses the same mechanism as the natural effect but is amplified by additional greenhouse gases from human activity."

Build the model yourself

Start in Natural mode: a modest amount of greenhouse gas, energy roughly in balance, temperature near +15 °C. Now push the slider up into Enhanced territory. Watch what happens to the longwave arrows escaping to space, to the energy-balance bar, and to the planet's temperature. Nothing about the mechanism changes — there are just more molecules doing the same job.

Analogies that actually help

◆ ANALOGY

The doona (the best one for KK04) Sleeping under one doona keeps you at a comfortable, stable temperature — that's the natural effect. Throw on three more doonas and you keep producing the same body heat, but less of it escapes, so you overheat — that's the enhanced effect. Same body, same heat source, more insulation. It captures the crucial point: nothing about the mechanism changed; there's just more of the blocking layer.

◆ ANALOGY

The car in the car park Sunlight (shortwave) pours through the glass and is absorbed by the seats, which re-radiate heat (longwave) the glass won't let back out. The cabin heats up. It's a handy picture of "shortwave in, longwave trapped." ⚠ Limitation: a real car (and a real glass greenhouse) heats mostly by stopping warm air from circulating away, not by re-radiation. The atmosphere has no "glass roof" — it warms by absorption and re-emission. Don't lean on this one in an explanation question.

◆ ANALOGY

The energy bank account Income (incoming sunlight) is fixed. The natural effect is a budget that balances — what comes in goes out, steady balance. The enhanced effect is like quietly reducing your spending (less heat leaving) while income stays the same: the balance climbs. The planet's "balance" is its temperature.

Exam traps that cost real marks

⚠ EXAM TRAPS

These are completely different problems. Ozone depletion is caused by CFCs destroying stratospheric ozone, letting more UV through. The greenhouse effect is about longwave radiation and surface warming. Mixing them up is an instant red flag to an examiner. Fix: Ozone hole = UV problem. Greenhouse = infrared/warming problem. Keep them in separate boxes. "Trap" is too vague and slightly wrong. Gases don't hold heat in a net — they absorb longwave radiation and re-emit it in all directions, including back toward the surface. Fix: Use "absorb and re-emit longwave (infrared) radiation." That phrase is the markable mechanism. Greenhouse gases are largely transparent to incoming shortwave sunlight and absorb outgoing longwave infrared. Swapping these reverses the physics. Fix: Shortwave in (sunlight), longwave out (re-radiated by the warmed surface). The gases act on the longwave. The natural effect is essential and beneficial. The harm comes from the enhanced effect. A question asking for "the difference" wants you to flag this value distinction. Fix: Natural = good/essential (+33 °C of habitability). Enhanced = the additional, harmful warming. Humans did not create it — it predates us by billions of years. Humans enhanced an existing effect by adding gases. Fix: Humans intensified a natural process; they did not invent it.

Decode the command word

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

POINT

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

Practice exam questions

★ PRACTICE & SELF-MARK

"It's when greenhouse gases trap heat in the atmosphere and make the Earth hot." The natural greenhouse effect is the warming of Earth's surface caused by naturally occurring greenhouse gases absorbing and re-emitting longwave radiation, which raises the global average temperature from about -18°C to $+15^{\circ}\text{C}$ and is essential for life.

- 1 mark — warming caused by (naturally occurring) greenhouse gases absorbing/re-emitting longwave (infrared) radiation.
- 1 mark — raises global average temperature (e.g. $\sim -18^{\circ}\text{C} \rightarrow \sim +15^{\circ}\text{C}$) / is essential for life / is a stable, balanced process.

★ PRACTICE & SELF-MARK

"The natural greenhouse effect is natural and the enhanced one is caused by humans burning fossil fuels and pollution. The enhanced one is making the Earth warmer and is bad for the environment." The natural greenhouse effect is warming produced by naturally occurring greenhouse gases absorbing and re-emitting outgoing longwave radiation, keeping the surface at a stable $\sim +15^{\circ}\text{C}$ and making Earth habitable. The enhanced greenhouse effect uses the same mechanism but is the additional warming caused by human activity (e.g. CO_2 from fossil fuels, CH_4 from agriculture) raising greenhouse-gas concentrations. Because more longwave radiation is absorbed and re-emitted downward, less heat escapes to space and the energy budget becomes imbalanced, raising global temperatures. The natural effect is essential and balanced, whereas the enhanced effect is an additional, harmful, human-driven intensification.

- 1 mark — natural effect: naturally occurring GHGs absorb/re-emit longwave radiation (stable, essential, $\sim +15^{\circ}\text{C}$).
- 1 mark — enhanced effect caused by human activities increasing GHG concentrations (named source).
- 1 mark — same mechanism, but more longwave absorbed/re-emitted $\rightarrow$ less escapes $\rightarrow$ energy imbalance.
- 1 mark — clear difference stated: natural = essential/balanced; enhanced = additional/harmful warming.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

It's when greenhouse gases trap heat in the atmosphere and make the Earth hot." Vague ("trap heat"), no mechanism, no figures, and "make the Earth hot" misses that the natural effect is beneficial and stable. Reads like the enhanced effect. 0–1 / 2.

✓ STRONG / MODEL ANSWER

The natural greenhouse effect is the warming of Earth's surface caused by naturally occurring greenhouse gases absorbing and re-emitting longwave radiation, which raises the global average temperature from about -18°C to $+15^{\circ}\text{C}$ and is essential for life. Names the cause (natural GHGs), the mechanism (absorb/re-emit longwave) and the outcome (the $-18 \rightarrow +15$ figures). Either the mechanism or the figures earns the second mark.

X WEAK ANSWER

The natural greenhouse effect is natural and the enhanced one is caused by humans burning fossil fuels and pollution. The enhanced one is making the Earth warmer and is bad for the environment." Identifies cause (natural vs human) but never mentions the shared mechanism, no figures, and "pollution" is too loose. Treats them as separate things rather than one intensifying the other. $\sim 2 / 4$.

✓ STRONG / MODEL ANSWER

The natural greenhouse effect is warming produced by naturally occurring greenhouse gases absorbing and re-emitting outgoing longwave radiation, keeping the surface at a stable $\sim +15^{\circ}\text{C}$ and making Earth habitable. The enhanced greenhouse effect uses the same mechanism but is the additional warming caused by human activity (e.g. CO_2 from fossil fuels, CH_4 from agriculture) raising greenhouse-gas concentrations. Because more longwave radiation is absorbed and re-emitted downward, less heat escapes to space and the energy budget becomes imbalanced, raising global temperatures. The natural effect is essential and balanced, whereas the enhanced effect is an additional, harmful

X WEAK ANSWER

The sun's rays come in and the greenhouse gases trap them so the heat can't get out and it warms up the Earth." No distinction between shortwave and longwave (the question demands it), "trap" instead of absorb/re-emit, and it implies the gases block incoming sunlight. $\sim 1 / 4$.

✓ STRONG / MODEL ANSWER

Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface, warming it. The surface then re-emits energy as longwave (infrared) radiation. Greenhouse gases in the atmosphere absorb this outgoing longwave radiation and re-emit it in all directions, including back toward the surface. This returned radiation warms the surface and lower atmosphere beyond what incoming sunlight alone would achieve.